Jatin

Analog Circuit Design Engineer

jatin@example.com • github.com/jatin-analog-git • linkedin.com/in/jatin

Summary

Analog IC design engineer with 3+ years of experience in OTA design, data converters, and bandgap references. Proficient in the gm/ID methodology for systematic transistor sizing and automated design-space exploration using MATLAB and Cadence Virtuoso. Skilled at taking circuits from specification to post-layout silicon.

Experience

Senior Analog Design Engineer — *2023 – Present*
Top Semiconductor Corp — *India*

- Led design of high-speed ADC front-end circuits in 28nm CMOS, meeting all target specifications on first silicon.
- Developed automated transistor sizing scripts using the gm/ID methodology, reducing design-cycle time by 60%.
- Achieved 20% power reduction in folded cascode OTA designs through systematic optimization of bias and sizing.
- Mentored two junior engineers on analog design principles and Cadence Virtuoso workflows.

Analog Design Engineer — *2021 – 2023*
Innovation Circuits Inc — *India*

- Designed folded cascode OTAs for sensor-interface front-ends with >60 dB DC gain and >100 MHz GBW in 180 nm TSMC.
- Implemented flicker and thermal noise co-optimization, meeting input-referred noise targets ($<5\,\mathrm{nV}/\sqrt{\mathrm{Hz}}$).
- Collaborated on multi-disciplinary IC projects spanning RF front-end, data converter, and power management blocks.

Analog Design Intern — *2020 – 2021*
Startup Electronics — *India*

- Assisted in bandgap reference design achieving 15 ppm/°C temperature coefficient with curvature compensation.
- Performed full PVT corner simulations and summarised results for tape-out design review.

Education

M.Tech in VLSI Design — *2018 – 2020*
Indian Institute of Technology — *India*

- Thesis: *High-Performance OTA Design Optimization via gm/ID Methodology* CGPA: 9.2/10
- Teaching Assistant for Analog IC Design (2 semesters).

B.Tech in Electronics & Communication Engineering — *2014 – 2018*
National Institute of Technology — *India*

- Graduated with Distinction. Final-year project: Low-Power 12-bit SAR ADC.

Selected Projects

Folded Cascode OTA Optimizer MATLAB, Cadence Virtuoso, 180nm TSMC — *2024 – 2025*

- Automated sizing tool using gm/ID lookup tables — inputs are gain, GBW, and power; output is a complete SPICE netlist.
- Post-layout: 72 dB DC gain, 156 MHz GBW, 68° phase margin, 0.82 mW.
- github.com/jatin-analog-git/ota-optimizer

12-bit SAR ADC 65nm CMOS, Cadence, HSPICE — *2023*

- Designed bootstrapped sampling switch and monotonic-switching capacitive DAC; targeting <1 mW at 10 MS/s.

Sub-1V Bandgap Reference 180nm TSMC, Spectre — *2022*

- Ultra-low-voltage bandgap reference with 15 ppm/°C TC and second-order curvature compensation.

Skills

Analog OTA Design · ADC/DAC · Bandgap · PLL
EDA Tools Cadence Virtuoso · HSPICE · Spectre · Calibre
Programming MATLAB · Python · SKILL · Verilog-AMS · Shell
Methods gm/ID · Monte Carlo · PVT · Noise Analysis · Layout
Other Git · LaTeX · Design Reviews

Awards & Honours

- **Performance Award** — Top Semiconductor Corp, 2024. Recognised for first-pass silicon success on high-speed ADC front-end in 28nm CMOS.
- **1st Place, IBM Qiskit Fall Fest 2025** — Daily Eigenchallenges (global IBM-hosted quantum computing competition).
- **Gold Medal, M.Tech** — Indian Institute of Technology, 2020. Awarded for 1st rank overall and best thesis in VLSI Design.
- **Gold Medal, B.Tech** — National Institute of Technology, 2018. Awarded for 1st rank overall and best thesis in Electronics & Communication Engineering.

Publications

- **Jatin** et al., "Automated gm/ID-Based Sizing Framework for Folded Cascode OTAs," *IEEE ISCAS*, 2024.
- **Jatin** et al., "Noise-Optimized OTA Design for Sensor Interfaces in 180nm CMOS," *IEEE TCAS-II*, 2023.
- **Jatin** & co-authors, "Low-Power SAR ADC with Monotonic Switching DAC in 65nm," *IEEE VLSI Symposium*, 2023.
- **Jatin** et al., "Sub-1V Bandgap Reference with Curvature Compensation," *IEEE ESSCIRC*, 2022.
- **Jatin** & co-authors, "gm/ID Characterization for 180nm TSMC PDK," *IEEE MWSCAS*, 2021.